Solutions - Week 6

Derivatives

1 A

2 D

3 Used by companies to minimise risk (as distinct from those *trading* in derivatives).

Options

Allow a company to protect itself against adverse movements in the finance environment, and retain ability to profit from favourable movements

Futures

'Locks in' the value of assets or liabilities

Guarantees the value of receipts or payments

Example:

–Company raises capital by borrowing at floating interest rate

–Can use interest rate futures to fund any increases in future interest payments (but will have to pay over any interest saved if market rates fall)

Swaps

Protect against adverse fluctuations in interest rates (or currency) where company has floating rate debt (or currency exposure)

4 Options give the right to buy or sell the underlying asset, whereas futures require completion of the transaction at the conclusion of the period. That makes each type of instrument suitable for dealing with a different type of risk. A future would be suitable for creating certainty as to the cost of a particular product in the future.

For example, a farmer might use the futures markets to sell a crop for a fixed price at a specified date. That would mean that the selling price was fixed and the farmer would be unaffected by a drop in price. However, the farmer could not benefit from any subsequent increase in the price because delivery would still have to be completed. An option to sell the product would leave the farmer free to exercise the option if the price fell below the strike price, but it would also be possible to let the option lapse and sell at a higher market price. In a sense, the premium paid for an option might be thought of as an insurance policy to protect against a specific downside risk, whereas a future protects against all risks, good and bad.

5. The size of the contract allows small investors to purchase it. The fact that the contracts can be traded 24 hours a day makes the contract more liquid and allows investors to speculate using current information from markets around the world. It also makes it more convenient for foreign investors to trade the contracts.

6. Hedgers are investors who own funds composed of stocks from the S&P 500; they will sell futures contracts to hedge against the risk that the market falls. Speculators are trying to profit from movements in the market; they will sell futures if they expect the market to fall, and buy futures if they expect the market to rise.

7. Because option buyers incur no obligations, their losses are limited to the price paid for the option. Their potential gains, however, can be large. Sellers must buy or sell the underlying asset at the strike price if the option is exercised, so their losses are unlimited. When writing a call option, the seller can lose money if the price of the underlying asset rises; however, if the seller owns the asset, then he or she is insured against these potential losses and issuing a call option is not very risky.

8. If the company has a comparative advantage in borrowing in the floating rate market, it can reduce its overall interest costs by borrowing at a floating interest rate and entering a swap agreement where it makes fixed payments and receives payments that fluctuate with the interest rate. The net effect is that its payments are fixed but, because it exploits its comparative advantage in the floating rate market where it can borrow relatively cheaply, the overall cost is lower than borrowing directly from the fixed rate market.